Linear systems and elimination

The exact fifth complete-pivoting growth factor

Difficulty: challenging **Importance:** interesting to specialist

Status: open.

Context and notation

All elimination in this problem is in exact arithmetic. Write $\|A\|_{\max} = \max_{ij} |a_{ij}|$. A pivoting path creates successive active Schur complements $S_1 = A, S_2, \dots, S_n$, with row/column permutations as appropriate. Its element-growth factor is

$$\rho(A) = \frac{\max_{1\le j\le n} \|S_j\|_{\max}}{\|A\|_{\max}}.$$

Partial pivoting chooses a largest-magnitude entry in the active first column; complete pivoting chooses one anywhere in the active matrix. If ties occur, a universal statement includes every admissible tie choice; a supremum includes all admissible paths. These conventions remove implementation-dependent ambiguity. Matrices are nonsingular unless otherwise stated.

Problem statement

Let

$$g_5 = \sup\{\rho_{\mathrm{CP}}(A) : A \in \mathbb{R}^{5\times 5} \text{ nonsingular}\}.$$

Let α be the unique real root in $(4, 5)$ of the integer polynomial P_5 displayed in Equation (2.15) of the primary reference below. This definition specifies an exact algebraic number; numerically $\alpha \approx 4.13251707863247285 42$. Prove or disprove

$$g_5 = \alpha.$$

All complete-pivoting paths are included in the supremum. The lower bound $g_5 \ge \alpha$ is already established in the cited paper. The unresolved part is global optimality over all allowable patterns of active constraints, not exact evaluation of a particular numerical example. Order five is independently singled out by the literature as the first unresolved dimension; this catalog does not split the same question into entries for every subsequent order.

Reference

Chen, Edelman, and Urschel, *The largest 5th pivot may be the root of a 61st degree polynomial*, February 2026, Equation (2.15), Conjecture 2.1, Theorem 2.2, and Theorem 3.5. The current rigorous interval is $\alpha \le g_5 \le 4.84$.

Status check

Searches for the paper title and *complete pivoting maximum five 2026* found no global proof. Shah–Urschel's August asymptotic result leaves this exact finite-dimensional optimum undetermined.